

HR Analytics Project — IBM Employee Attrition

Personal project notes & data reference

Why I picked this project

I wanted a portfolio project that actually felt like real analyst work, not just a tutorial. Employee attrition is a problem every company cares about, and the IBM HR dataset gave me enough columns to practice the full workflow — cleaning data in SQL, finding patterns through EDA, and presenting findings in a dashboard a manager could actually use.

About the dataset

1,470 employee records, 35 columns. Covers basic demographics (age, gender, marital status), job details (department, role, level), pay (monthly income, daily rate, etc.), satisfaction scores, and tenure (years at company, years since promotion). The target column is **Attrition** — Yes or No, whether that employee left.

How I cleaned the data

I checked for nulls and duplicates first — found none, so the dataset was already fairly clean. I dropped four columns that had zero variation and added no value: EmployeeCount, EmployeeNumber, Over18, and StandardHours. Every employee had the same value in those, so keeping them would just add noise.

Columns I had to think through carefully

A few columns look similar but mean very different things, and I made sure I understood the difference before using them:

MonthlyIncome vs MonthlyRate. MonthlyIncome is the real salary an employee actually receives. MonthlyRate is an internal number IBM uses for budgeting — the employee never sees it. I only used MonthlyIncome for any salary-related analysis.

YearsAtCompany vs TotalWorkingYears. YearsAtCompany measures loyalty to IBM specifically. TotalWorkingYears is the employee's entire career across every company they've worked for. I used YearsAtCompany when looking at retention specifically tied to IBM.

HourlyRate / DailyRate. Both are internal reference numbers (used for overtime calculations and project costing), not actual pay. I treated these as lower priority for analysis.

What I found in SQL

I wrote a set of queries in PostgreSQL to break attrition down by department, gender, age group, income band, overtime status, and tenure. The overall attrition rate came out to **16.12%**, which is above the healthy industry range of 10-12%. A few numbers stood out enough that I kept coming back to them:

- Sales had the highest department attrition at 20.63% — about 1 in 5 Sales employees leaves.
- Employees working overtime left at almost 3x the rate of those who didn't (30% vs 10.44%).
- Younger employees (18-25) had a 35.77% attrition rate — by far the highest age group.
- Sales Representatives were the single highest-risk job role at 39% attrition.

The finding that surprised me most

Individual factors were interesting, but combining them told a much sharper story. When I filtered for employees aged 18-25, working overtime, AND earning a low salary, the attrition rate jumped to **73%**. That's roughly 3 out of every 4 people in that exact situation leaving. It made it clear that attrition isn't really about one cause — it's usually a stack of pressures hitting the same person at once.

I also noticed something counter-intuitive: employees who had just been promoted showed the *highest* attrition, not the lowest. My read on this is that promotions at IBM often come with a new title but not enough of a pay increase, so people use the promotion to look for a better-paying role elsewhere instead of staying.

KPIs I tracked on the dashboard

Metric	Value
Total Employees	1,470
Attrition Count	237
Attrition Rate	16.12%
Average Monthly Income	\$6,503
Average Tenure	7 years
Average Job Satisfaction	3 / 4

How the dashboard is structured

I split the Power BI dashboard into five pages, each answering a different question:

- Executive Summary** — The big-picture numbers — total employees, attrition rate, department and gender breakdowns. This is the page I'd open first in an interview.
- Attrition Analysis** — The 'why' page — overtime, income, tenure, promotion gap, business travel, all charted against attrition.

Employee Demographics — Who actually works at IBM — gender split, marital status, education levels, job roles, age groups.

Job Satisfaction & Performance — How satisfaction, work-life balance, training frequency, and performance ratings connect to attrition.

Recommendations — My actual suggestions for HR, plus a rough estimate of the cost savings if attrition improves.

What I'd recommend to HR

Review pay for low earners. Anyone under \$3,000/month showed noticeably higher attrition — even a modest increase here could help.

Cap overtime. No employee should be doing overtime more than 2 days a week on a regular basis.

Build a program for younger employees. Mentorship and a visible career path matter more than people assume at this stage.

Pair promotions with real pay increases. A title change alone isn't retaining people.

Look closely at Sales. High travel, high pressure, and the highest attrition of any department — this needs department-specific attention, not a company-wide fix.

If IBM brought attrition down from 16.12% to around 10% (the industry benchmark), that's roughly 89 fewer people leaving each year. Using the standard estimate that replacing an employee costs 30-50% of their annual salary, that works out to an estimated **\$2-4 million in savings annually**.

Tools used

PostgreSQL (pgAdmin) for cleaning and querying the data. Power BI Desktop for the dashboard — Power Query for the data transformations, DAX for the measures, and a mix of bar, column, donut, line, area, and funnel charts depending on what each chart needed to show.